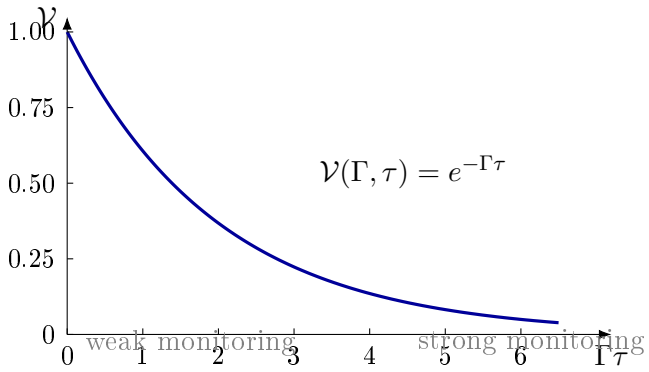


Figure 1: Fringe visibility vs. which-way coupling in SST. The parameter Γ encodes EM impedance to the $\mathcal{R}$ phase (path information rate), τ is the transit time. Exponential decay of visibility reflects coherence loss prior to recombination.

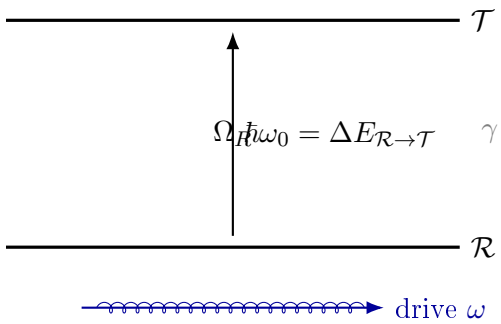
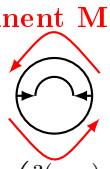


Figure 2: Two-level SST manifold for $\mathcal{R} \leftrightarrow \mathcal{T}$ with Rabi drive. At resonance $\omega = \omega_0$, the knotting probability oscillates at Ω_R and decays at rate γ, enabling pump–probe control of collapse within an interferometer.

TikZ Sketch: Dipole Fields

Permanent Magnets



$$\mathbf{B} = \frac{\mu_0}{4\pi}\left(\frac{3(\mathbf{m}\cdot\mathbf{r})\mathbf{r}}{r^5} - \frac{\mathbf{m}}{r^3}\right)$$

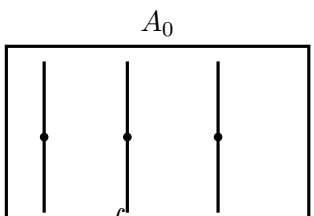
$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$$

Permanent Electrets

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\left(\frac{3(\mathbf{p}\cdot\mathbf{r})\mathbf{r}}{r^5} - \frac{\mathbf{p}}{r^3}\right)$$

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}$$

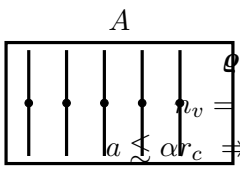
TikZ Sketch: Fixed Swirl Flux, Shrinking Area ⇒ Vortex Nucleation



$$\Phi_{sw} = \int_{A_0} \boldsymbol{\varrho}\cdot d\mathbf{A} = N\kappa$$

(fixed)

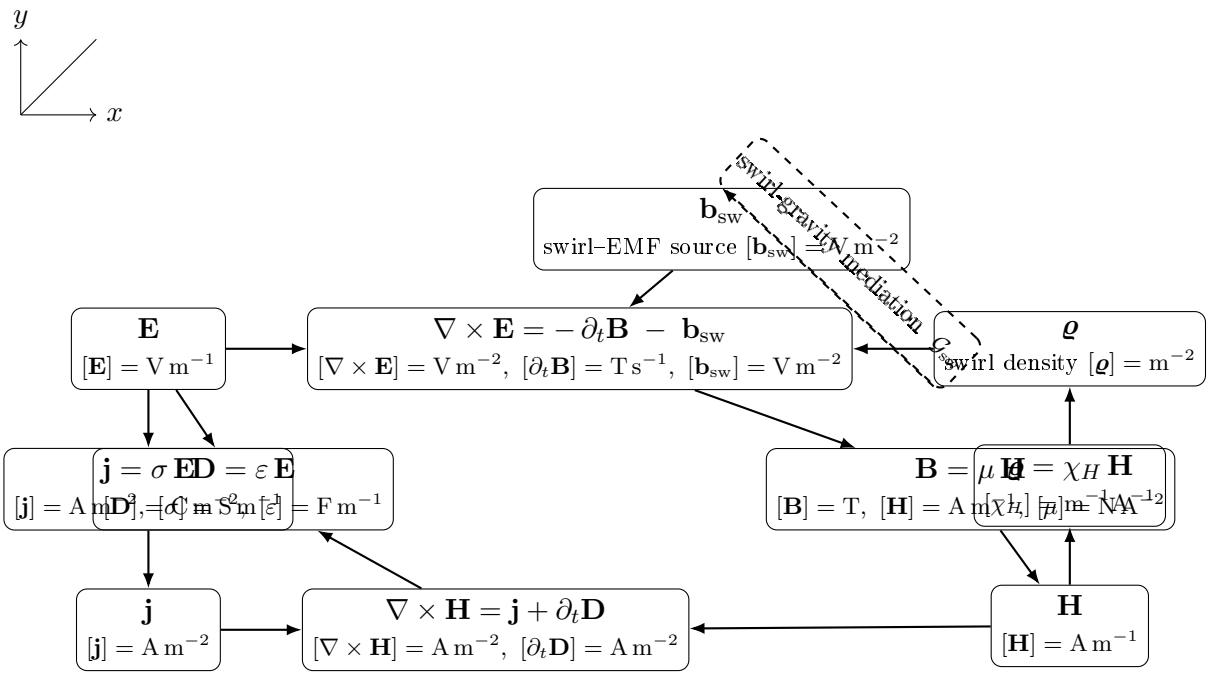
shrink →



$$\boldsymbol{\varrho}(A) = \frac{N\kappa}{A}\,\hat{\mathbf{n}}$$

$$n_v = \frac{N}{A}, \quad a \sim n_v^{-1/2}$$

$$a \lesssim \alpha r_c \Rightarrow \text{vortex lattice onset}$$


y
x
$\mathbf{b}_{\mathrm{sw}}$
swirl–EMF source $[\mathbf{b}_{\mathrm{sw}}] = \mathrm{N\,m^{-2}}$
swirl-gravity mediation
$\mathbf{E}$
$[\mathbf{E}] = \mathrm{V\,m^{-1}}$
$\nabla \times \mathbf{E} = -\,\partial_t \mathbf{B} \;-\; \mathbf{b}_{\mathrm{sw}}$
$[\nabla \times \mathbf{E}] = \mathrm{V\,m^{-2}},\ [\partial_t \mathbf{B}] = \mathrm{T\,s^{-1}},\ [\mathbf{b}_{\mathrm{sw}}] = \mathrm{V\,m^{-2}}$
$\boldsymbol{\varrho}$
swirl density $[\boldsymbol{\varrho}] = \mathrm{m^{-2}}$
$\mathbf{j} = \sigma\,\mathbf{E}$
$[\mathbf{j}] = \mathrm{A\,m^{-2}},\ [\sigma] = \mathrm{S\,m^{-1}}$
$\mathbf{D} = \varepsilon\,\mathbf{E}$
$[\mathbf{D}] = \mathrm{C\,m^{-2}},\ [\varepsilon] = \mathrm{F\,m^{-1}}$
$\mathbf{B} = \mu\,\mathbf{H}$
$[\mathbf{B}] = \mathrm{T},\ [\mathbf{H}] = \mathrm{A\,m^{-1}},\ [\mu] = \mathrm{N\,A^{-2}}$
$\mathbf{M} = \chi_H\,\mathbf{H}$
$[\chi_H] = 1$
$\mathbf{j}$
$[\mathbf{j}] = \mathrm{A\,m^{-2}}$
$\nabla \times \mathbf{H} = \mathbf{j} + \partial_t \mathbf{D}$
$[\nabla \times \mathbf{H}] = \mathrm{A\,m^{-2}},\ [\partial_t \mathbf{D}] = \mathrm{A\,m^{-2}}$
$\mathbf{H}$
$[\mathbf{H}] = \mathrm{A\,m^{-1}}$